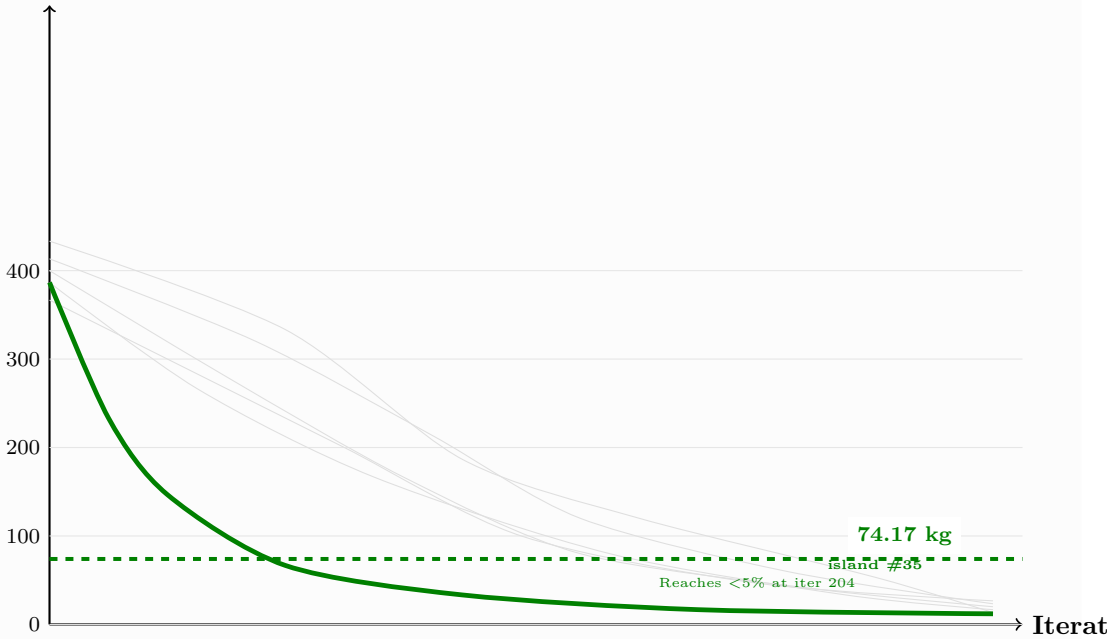


V6.2 Optimization Landscape

1. Convergence: 58/60 Islands → 74.17 kg

Grey = individual islands. Green = best island (#35).

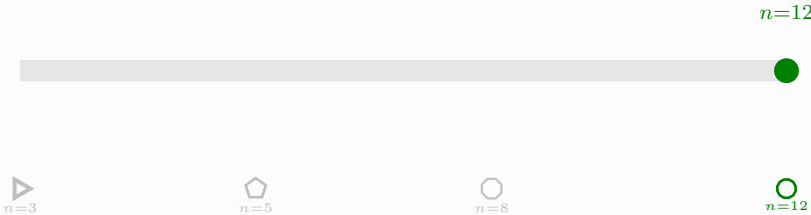
Best mass (kg)



Bounds: $n \in [3, 12]$, $r_{hub} \in [1, 10]m$, $r_{bot} \in [0.1, 5]m$,
 $\beta \in [-0.8, 0.8]$, $n_{rings} \in [5, 16]$, $n_{exp} \in [0, 6]$, $bank \in [0, 60]^\circ$

2. Polygon Search: Converged to $n=12$

Optimizer freely varied $n \in [3, 12]$ alongside 10 other parameters



Higher $n \rightarrow$ thinner beams $\rightarrow$ smaller coupled knuckles

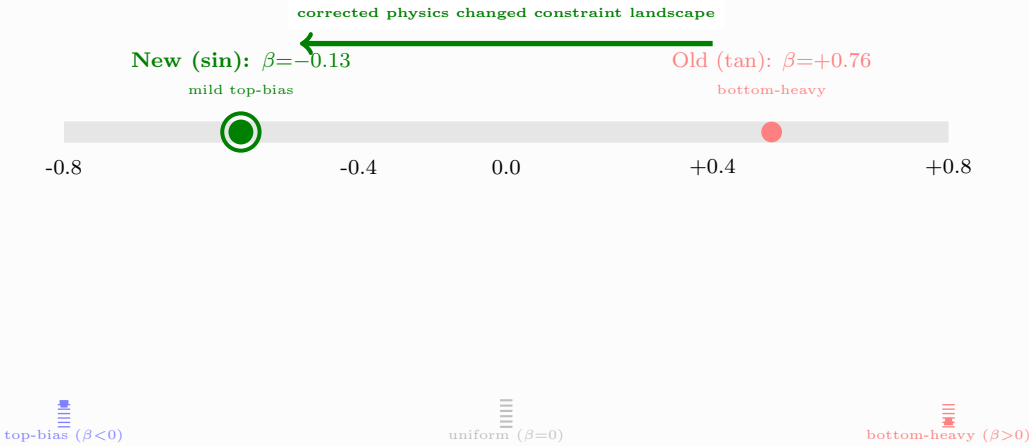
At $n=12$, per-beam compression is $N/12$ – single hub rotor suffices

Single-parameter sweep infeasible. 58/60 islands independently converged to $n=12$.

3. Density Profile: β Sign Flip ($+0.76 \rightarrow -0.13$)

The tan formula created different constraints – this is physics evolution, not error correction. $n_{exp} \in [0, 6]$ – expansion blades inherit span, chord, and count from main BEM rotor. Only n_{exp} and bank angle are free.

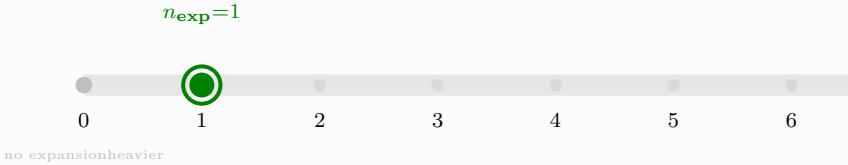
The old tan formula overstated beam compression, making buckling the dominant constraint – rings HAD to cluster at the bottom ($\beta=+0.76$) to resist buckling. The corrected sin formula showed real compression is lower. At $n=12$ with a hub expansion rotor, buckling relaxes and the radial force gradient from the expansion rotor favours rings near the hub ($\beta=-0.13$).



Extreme sensitivity: even $\beta=-0.12857$ (5 decimal places) fails the FoS check. Only the full 17-digit campaign value (-0.12856962561009427) works. The optimum is a razor-sharp constraint intersection – not a smooth valley but a needle in a constraint haystack.

4. Expansion Stations: $n_{exp}=1$ Optimal

Optimizer freely varied $n_{exp} \in [0, 6]$ alongside 10 other parameters



Each extra station adds 3 blades + knuckle hardware + parasitic drag.
At $n=12$, per-beam compression is low enough that one hub rotor provides sufficient F_{radial} for the entire shaft.
The old V6 at $n=8$ needed 3 rotors because higher per-beam compression required more radial spreading.

Limitation: expansion blades use the same mould as the main rotor.
Blade span was not a free parameter – the optimizer could not try shorter or longer expansion blades.

Single-parameter sweep infeasible. Optimizer freely explored 11-D space. $n_{exp}=0$ (no expansion) is heavier – some radial spreading is required for feasibility. Blade length remains an unexplored degree of freedom.